Ingenitex (BD) Limited

Jamgora , Ashulia, Dhaka
(23.93855233203, 90.2872665048)
13 February 2023



RSC RMG SUSTAINABILITY COUNCIL

Building Information

1. Factory Building is a six storied (G+5) reinforced concrete (RC) building

Observations

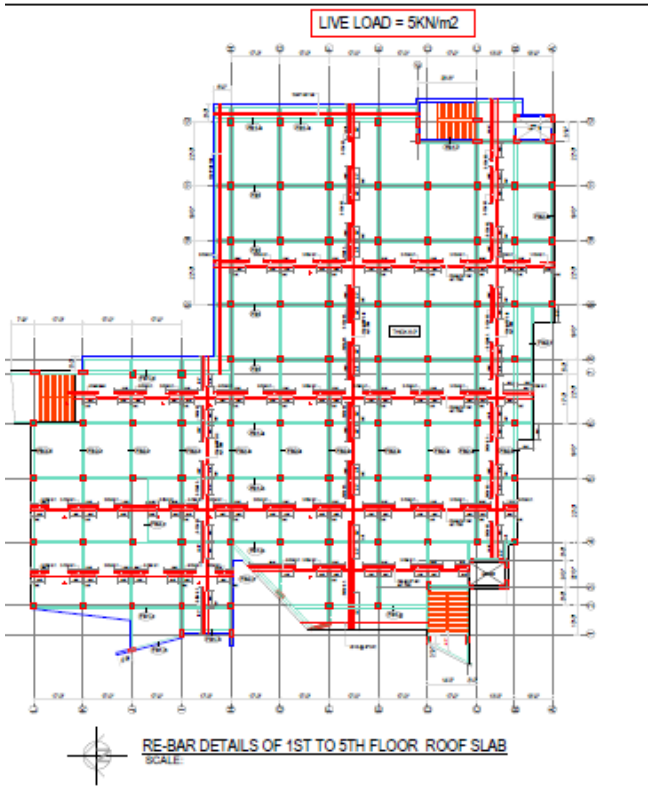
Inconsistences in design report

5.4.2 Adequacy Check of Beam

1. Grade Beam Adequacy Check:

Label	Story	Section	Location	(-) Moment kip-ft	(-) Combo	As Top in²	Provided top reinforcement (in2)	remarks	(+) Moment kip-ft	(+) Combo	As Bot in²	provided bottom reinforcement (in2)	remarks
B70	GF	GB-1_12X24	End-J	-90.7677	RCC-17	1.0062	1.47	OK	73.4044	RCC-26	0.868	1.47	OK
B71	GF	GB-1_12X24	End-I	-85.608	RCC-26	0.868	1.47	OK	69.6448	RCC-25	0.868	1.47	OK
B71	GF	GB-1_12X24	Middle	-32.7635	RCC-17	0.4926	1.47	OK	36.2715	RCC-17	0.5702	1.47	OK
B71	GF	GB-1_12X24	End-J	-88.513	RCC-17	0.9468	1.47	OK	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
B85	GF	GB-1_12X24	End-I	-106.1119	RCC-26	1.1917	1.47	OK	143.1565	RCC-21	1.5288	1.47	Say OK
B85	GF	GB-1_12X24	Middle	-66.7689	RCC-25	0.6258	1.47	OK	53.3806	RCC-26	0.7524	1.47	OK
B85	GF	GB-1_12X24	End-J	-142.8385	RCC-25	1.5288	2	OK	135.424	RCC-26	1.3956	1.47	OK
B86	GF	GB-1_12X24	End-I	-77.3563	RCC-16	1.182	1.47	OK	36.3632	RCC-24	0.868	1.47	OK
B86	GF	GB-1_12X24	Middle	-19.3561	RCC-16	0.1759	1.47	OK	25.1774	RCC-24	0.868	1.47	OK
B86	GF	GB-1_12X24	End-J	-69.8829	RCC-26	0.868	1.47	OK	19.3341	RCC-16	0.1759	1.47	OK
B88	GF	GB-1_12X24	End-I	0	RCC-15	0.0262	1.47	OK	1.1366	RCC-15	0.0464	1.47	OK
B88	GF	GB-1_12X24	Middle	-38.1763	RCC-15	0.5613	1.47	OK	20.3561	RCC-15	0.3319	1.47	OK
B88	GF	GB-1_12X24	End-J	-81.5843	RCC-15	0.8869	1.47	OK	40.7921	RCC-15	0.5962	1.47	OK
B92	GF	GB-1_12X24	End-I	-80.758	RCC-24	0.868	1.47	OK	70.7235	RCC-23	0.868	1.47	OK
B92	GF	GB-1_12X24	Middle	-36.0818	RCC-23	0.4564	1.47	OK	39.7108	RCC-24	0.5666	1.47	OK
B92	GF	GB-1_12X24	End-J	-79.0962	RCC-23	0.868	1.47	OK	72.4401	RCC-24	0.868	1.47	OK
B93	GF	GB-1_12X24	End-I	-78.1968	RCC-24	0.868	1.47	OK	67.2995	RCC-23	0.868	1.47	OK
B93	GF	GB-1_12X24	Middle	-35.4263	RCC-24	0.5183	1.47	OK	37.3038	RCC-16	0.1529	1.47	OK
B93	GF	GB-1_12X24	End-J	-74.0905	RCC-23	0.868	1.47	OK	68.3056	RCC-24	0.868	1.47	OK
B97	GF	GB-1_12X24	End-I	-95.7512	RCC-24	1.0092	1.47	OK	85.8797	RCC-23	0.868	1.47	OK
B97	GF	GB-1_12X24	Middle	-28.3705	RCC-24	0.3779	1.47	OK	35.435	RCC-24	0.4773	1.47	OK
B97	GF	GB-1_12X24	End-J	-91.7807	RCC-23	0.9049	1.47	OK	95.6653	RCC-24	1.0088	1.47	OK
B90	GF	GB-1_12X24	End-I	-85.1373	RCC-18	0.9584	1.47	OK	64.0404	RCC-25	0.868	1.47	OK
B90	GF	GB-1_12X24	Middle	-31.2202	RCC-17	0.4984	1.47	OK	36.015	RCC-17	0.5657	1.47	OK
B90	GF	GB-1_12X24	End-J	-94.5654	RCC-17	1.0561	1.47	OK	76.5446	RCC-26	0.868	1.47	OK
B91	GF	GB-1_12X24	End-I	-69.4577	RCC-24	0.868	1.47	OK	36.2545	RCC-16	0.1253	1.47	OK
B91	GF	GB-1_12X24	Middle	-31.1075	RCC-16	0.2714	1.47	OK	32.5342	RCC-16	0.4753	1.47	OK
B91	GF	GB-1_12X24	End-J	-69.5015	RCC-23	0.868	1.47	OK	32.5726	RCC-24	0.4753	1.47	OK
B92	GF	GB-1_12X24	End-I	-111.7662	RCC-26	1.3029	1.47	OK	110.1437	RCC-25	1.1795	1.47	OK
B92	GF	GB-1_12X24	Middle	-47.3424	RCC-25	0.6552	1.47	OK	48.3472	RCC-18	0.4934	1.47	OK
B92	GF	GB-1_12X24	End-J	-116.3885	RCC-25	1.2438	1.47	OK	106.7019	RCC-26	1.1466	1.47	OK
B95	GF	GB-1_12X24	End-I	-87.4334	RCC-16	0.902	1.47	OK	68.099	RCC-23	0.868	1.47	OK
B95	GF	GB-1_12X24	Middle	-26.4092	RCC-23	0.3763	1.47	OK	33.8836	RCC-23	0.4785	1.47	OK
B95	GF	GB-1_12X24	End-J	-83.1291	RCC-23	0.8875	1.47	OK	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
B96	GF	GB-1_12X24	End-I	-129.6652	RCC-24	1.7164	1.47	OK	120.9595	RCC-23	1.5756	1.47	Say OK
B96	GF	GB-1_12X24	Middle	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
B96	GF	GB-1_12X24	End-J	-120.5631	RCC-23	1.5711	1.47	Say OK	120.2747	RCC-24	1.0089	1.47	OK

Some of the grade beam seems less reinforcement than the required. Building engineer is required to check design, load and flexural capacity of the beams.



Slab adequacy not included in the design report.

Observation: Factory Building

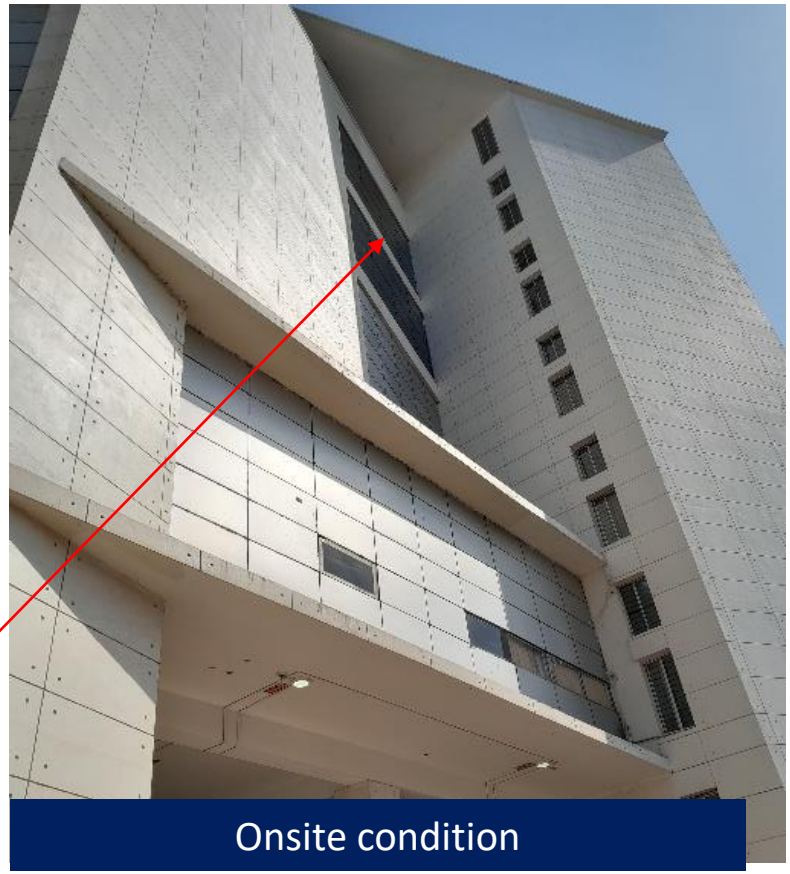
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Discrepancies in the as-built drawing

Observation: Factory Building



Architectural floor layout 3rd floor to roof floor



Onsite condition

Marked portion floor slab has not constructed on 3rd to 5th floor. So that, beam was found diagonally connected with column to column. But in as-built drawing slab and straight beam were shown. Building engineer is required to revise the as built drawing.

Observation: Factory Building



Floor layout plan



350 mm toilet buildup not shown in architectural drawing

Observation: Factory Building

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Hairline crack on beams and crack in the brick walls

Observation: Factory Building



Hairline crack on beam



Crack on brick wall and column joint

Hairline crack on beam and crack on brick wall and column joint observed. Building engineer is required to investigate the reason of crack and repair with suitable method.

Observation: Factory Building

Anchorage and bracing of storage racks and water tank







Nonstructural elements like storage racks were not anchored properly as per seismic provision of BNBC. Building engineer is required to provide the proper anchorage as per BNBC seismic requirements.

Observation: Factory Building

Complete the water/heat proofing works on the roof

Observation: Factory Building



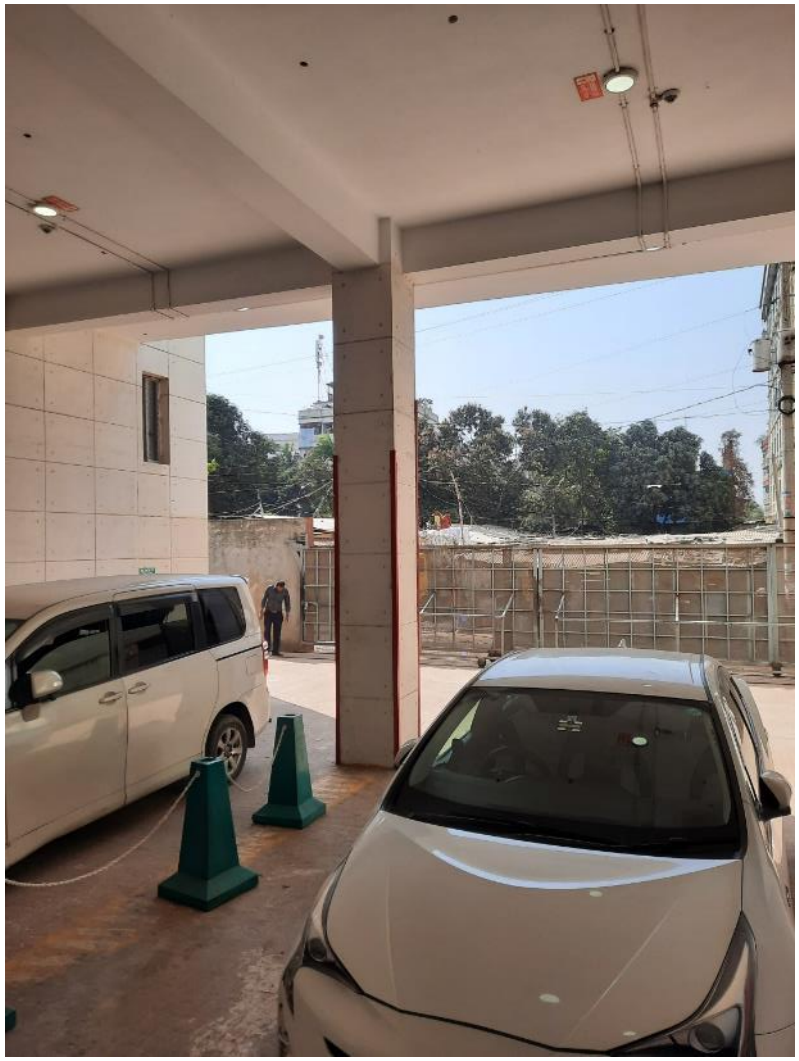
Incomplete water/heat proofing works observed on the roof. Building engineer is required to complete the water/heat proofing works on the remaining part of roof.

Observation: Factory Building

Inadequate vehicle impact protection system

Observation: Factory Building





The vehicle impact protection of ground floor columns appeared to be inadequate for heavy trucks/lorry. Building engineer is required to design and construct a proper vehicle impact protection system around the susceptible columns on ground floor.

Observation: Factory Building

Tests carried out



Stone Chips

Tests carried out

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Priority Actions

Problems Observed

Item 1: Inconsistences in design report.

Item 2: Discrepancies in as-built drawings.

Item 3: Hairline crack on beams and crack in the brick walls.

Item 4: Anchorage and bracing of storage racks and water tank

Item 5: Complete the water/heat proofing works on the roof.

Item 6: Inadequate vehicle impact protection system.

Item No.	Observation	Recommended Action Plan	Recommended Timeline
01	Inconsistences in design report.	Building engineer is required to design, load and flexural capacity of the beams.	6-weeks
02	Inconsistences in design report.	Building engineer is required to revise the design report as per BNBC 1.9.1 requirements.	6-weeks
03	Inconsistences in design report.	Carry out remedial works where required.	6-months
04	Discrepancies in as-built drawings.	Building engineer is required to survey the whole structure and prepare accurate as-built drawings.	6-weeks
05	Hairline crack on beams and crack in the brick walls.	Building engineer is required to investigate the reason of crack and repair with suitable method.	6-weeks

Item No.	Observation	Recommended Action Plan	Recommended Timeline
06	Anchorage and bracing of storage racks.	Building engineer is required to provide the proper anchorage as per BNBC seismic requirements.	6-weeks
07	Complete the water/heat proofing works on the roof.	Building engineer is required to complete the water/heat proofing works on the remaining part roof.	6-weeks
08	Inadequate vehicle impact protection system.	Building engineer is required to design and construct a proper vehicle impact protection system around the susceptible columns on ground floor.	6-weeks